GMI **German-Malaysian Institute**

DIPLOMA PROGRAMME

MINI PROJECT

Academic Period: JANUARY 2026

Code	**CBS 2343**	**Course Name**	**Wireless Network Security**
Title			
Name	**1.**	**ID**	**1.**
	2.		**2.**
	3.		**3.**
	4.		**4.**
Program/ Group		**Examiner**	**NUR DAYANA AHMAD DAUD**

No	**Assessment Criteria**	**CLO**	**PO**	**Level of Diff**	**Full Marks**	**Score**	**Marks**	**Comment**
1.	All criteria are based on the rubric provided.	2, 3	2, 6	P3, A3	40			
2								
3.								
4.								
5								
				Total				
				Total Marks	/40			
	Total	**PO**			/40			
				Signature				

QUESTION:

In this project, you are required to build a mini-project for wireless networks to detect and monitor malicious activities and unauthorised access attempts. Wireless networks are more susceptible to attacks due to their nature of transmission, and this project will help students identify potential security breaches or anomalies in wireless networks.

You are required to research current technologies and tools to use for this project. The following are suggestions:

1. Python
2. Wireshark
3. Kali Linux
4. User Interface: You can create a simple GUI to visualize detected attacks and network statistics using any suitable tools.

Rules and Regulations:

1. Each group may have a maximum of 4 members.
2. Each group must complete the project by the assigned deadline, before the end of TW18.
3. All group members must actively participate and contribute to the project.
4. Ensure that all sources used in the project are properly cited.
5. Each group must present their findings clearly.
6. Late submissions will be penalized according to the project guidelines.

WIRELESS NETWORK SECURTY MINI PROJECT RUBRIC

NO.	CRITERIA	EXCELLENT (4)	GOOD (3)	FAIR (2)	NEED IMPROVEMENT (1)	SCORE
	GROUP COMPONENT (TOTAL MARK = 28)					
1	Project Scope & Relevance	Clearly defines a relevant, impactful, and well-focused project related to wireless network security.	Defines a relevant project with some minor gaps in scope or relevance.	The project is somewhat relevant but lacks depth in application or focus.	The project lacks relevance or is not directly related to wireless network security.	/4
2	Technical Implementation	Implements a robust solution with all required functionality, working flawlessly across different environments.	Implements most of the functionality correctly; minor issues may exist but are resolved.	Implements basic functionality but encounters significant bugs or lacks key features.	The implementation is incomplete or major functionalities do not work.	/4
3	Creativity & Innovation	Demonstrates a high degree of creativity, offering new insights, ideas, or methods to solve the problem.	Demonstrates some creativity with clear problem-solving techniques and improvements.	Offers a functional but fairly standard solution with little originality.	Shows minimal creativity and uses only basic, standard approaches without innovation.	/4
4	User Interface / Usability	Highly intuitive, well-organized, and easy to use interface (if applicable), with appropriate error handling.	Clear and functional interface, but could be improved for better user experience.	Interface works but is basic or lacks some essential usability features.	Interface is hard to use, lacks clarity, or is very basic.	/4
5	Security Features Implemented	Excellent implementation of security measures **(encryption, authentication, attack detection)** with detailed considerations.	Good implementation of security features but some aspects are underdeveloped or partially implemented.	Basic security features included but lacks depth or proper implementation.	Limited or no security measures implemented.	/4
6	Code Quality & Documentation	Code is clean, well-commented, and **easy to understand**. Documentation is thorough, clear, and well-structured.	Code is well-organized with adequate comments. Documentation covers key aspects but may lack some detail.	Code is somewhat disorganized or under-commented. Documentation is minimal or unclear.	Code is difficult to understand or lacks essential documentation.	/4
7	Testing & Validation	Thorough testing across different scenarios; all edge cases are handled well. Clear evidence of robust validation.	Testing is done in key scenarios; some edge cases may be missing or not fully validated.	Testing is limited, with major gaps in coverage or unhandled edge cases.	No or insufficient testing; results are unreliable or incomplete.	/4
	INDIVIDUAL COMPONENT (TOTAL MARK = 12)					
8	Understanding of Wireless Security Concepts	Clearly demonstrates in-depth understanding of wireless network security principles, threats, and mitigation strategies.	Demonstrates good understanding but with some gaps in advanced security concepts.	Basic understanding of wireless security, though some key concepts are missing.	Shows limited or incorrect understanding of wireless network security.	/4
9	**Presentation & Communication**	The presentation is clear, engaging, and demonstrates deep knowledge. Effective communication of the project's purpose and results.	The presentation is clear and covers the project's purpose and results effectively, though some areas could be explained better.	The presentation is understandable but lacks clarity or depth in explaining key aspects.	The presentation is unclear, disorganized, or does not communicate the project's purpose or results well.	/4
10	Documentation & Report	Report is comprehensive, clear, well-structured, and includes all relevant technical details, explanations, and future recommendations.	Report is clear and covers most technical aspects, with minor gaps or missing details.	Report is basic or incomplete, missing key details, explanations, or recommendations.	Report is poorly written, lacks structure, or is missing essential content.	/4
					TOTAL	**/40**

EVALUATION CRITERIA	**INDIVIDUAL COMPONENT (TOTAL MARK = 12)**			
	STUDENT 1	STUDENT 2	STUDENT 3	STUDENT 4
Understanding of project concepts				
Presentation & Communication				
Documentation & Report				

STUDENT'S NAME	**INDIVIDUAL MARK**	**GROUP MARK**	**TOTAL MARK**
STUDENT 1			
STUDENT 2			
STUDENT 3			
STUDENT 4			

YOUR REPORT BEGINS HERE

DOCUMENTATION & REPORT

INTRODUCTION

1. **Purpose of the Project:** Explain the main objective of the project, what it aims to achieve, and why it is important.
2. **Background/Context:** Provide background information that helps the reader understand the topic. This may include previous research, studies, or why the topic is significant.
3. **Scope and Focus:** Outline the scope of the project. What specific areas will you focus on?
4. **Methodology:** Briefly describe the tools that will be used to carry out the project.
5. **Significance of the Project:** Conclude by explaining the importance of the project. Why is this topic worth studying, and how will it contribute to the field or community?

CONTENTS

1. **Implementation Results**

    a. Coding

    b. Testing Results

    c. Photos of Implementation Testing

2. **Security Features**

    a. Encryption

    b. Authentication and Attack Detection

    c. User Interface / Usability

3. **Understanding of Wireless Security Concepts**

CONCLUSION

REFERENCES